SHARP TAILS, RECURRENT EGFS, AND EXACT FIBRES FOR ODD-SIDE LEAST-NEIGHBOUR WALKS ON LABELLED TREES

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Abstract. Fix a labelled tree and move a distinguished root to the least-labelled neighbour whose side of the incident edge has odd order; hold when no such neighbour exists. We solve this deterministic finite dynamics uniformly in the parity of the tree order. For odd order, every edge points in one direction and all recurrent states are fixed; for even order, the odd-cut edges form a forest of odd degrees and every cycle has length two. In both cases the sharp maximum tail is $\lfloor (n-1)/2 \rfloor$, with explicit witnesses. We give exponential generating functions for all odd-order fixed rooted trees and all even-order recurrent rooted trees. The latter uses an integrated odd-branch marker and remains valid even though one odd-cut component may contain several attracting two-cycles. Finally, a local necessary-and-sufficient predecessor test gives every labelled fibre and the sharp maximum fibre: $(n+1)/2$ for odd n and $n-1$ for even n. Exhaustion through $n = 8$ checks 2,223,278 state transitions. Classical Cayley/species facts receive zero contribution credit, and external status is `OWNER_AMBER/HOLD_EXTERNAL`.

1. Definition and ownership boundary

Let T be a tree on $[n] = \{1, \dots, n\}$ and let r be a distinguished vertex. For $v \sim r$, write $C_r(v)$ for the component containing v in $T - rv$, and call v eligible from r when $|C_r(v)|$ is odd. Define

$$F(T, r) = \begin{cases} (T, \min\{v \sim r : |C_r(v)| \text{ is odd}\}), & \text{if this set is nonempty,} \\ (T, r), & \text{otherwise.} \end{cases} \tag{1}$$

Labels are compared only after parity eligibility is determined. The underlying tree never changes.

Cayley's formula, rooted-tree exponential generating functions, and labelled SET constructions are classical [1–4]. They receive zero contribution credit. A bounded owner search found parity-edge algorithms and general deterministic tree walks, but no literal least-neighbour system (1) with the theorem package below. This non-hit establishes neither novelty nor priority.

Internally, the fixed-carrier marker walk is distinct from P114 forest pruning, P120 odd-fringe mirroring, P123 odd-component complementation, the killed Catalan parity rotation from the P122–P126 scout, P144 tree rotations, P148 level contraction, and P159 parallel odd-vertex pruning. P123 changes all edges inside selected odd connected components and obtains its clock from a parity-pruned split tree; P159 deletes every current odd-degree vertex and uses a binary-rank transfer inverse. In contrast, P195 never changes the tree: it moves one marker by odd edge-side sizes and least labels. The shared $\lfloor (n-1)/2 \rfloor$ scale, fixed/two-cycle silhouette, labelled EGF/species calculus, zeta conversion, and generic local-fibre language earn no separation or contribution credit. The paper claims only the specific orbit geometry, integrated least-label recurrent census, and incident-edge inverse atlas of (1).

2. Parity geometry and recurrent states

For even n, let $H(T)$ be the spanning subgraph consisting of edges whose deletion leaves two odd components.

Theorem 2.1 (complete recurrent classification). *If n is odd, orient every edge toward its odd side. The map follows the least-labelled outgoing neighbour, and its recurrent states are precisely the sinks, equivalently the roots whose incident branches all have even order. Every recurrent state is fixed.*

If n is even, eligibility is symmetric and the map is the least-neighbour map on the forest $H(T)$. Every vertex has odd degree in $H(T)$, so there are no fixed states, and every recurrent state belongs to a two-cycle $u \leftrightarrow v$ in which u and v are mutual least neighbours in $H(T)$.

Proof. When n is odd, the two sides of each edge have opposite parity, so exactly one orientation is present. A directed cycle would give an undirected cycle in T; hence an orbit terminates at a sink. The sink condition is exactly the asserted branch condition.

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When n is even, the two sides have the same parity. At a vertex v, the sum of all branch orders is $n-1$, which is odd; therefore an odd number of branches are odd and $\deg_H(v)$ is odd. The update is thus everywhere defined and is exactly least-neighbour motion in H. A directed cycle of length at least three would create an undirected cycle in the forest H; the only possible cycles are reciprocal edges. ☐

Remark 2.2 (essential nonuniqueness warning). A connected component of $H(T)$ need not contain a unique mutual-minimum edge. For example, a suitably labelled six-vertex double star has two. Theorem 2.1 asserts uniqueness only for the eventual cycle in each *functional-graph* component. None of the counts below assumes one cycle per component of H.

3. The sharp pointwise clock

Let $\tau(T,r)$ be the first time at which the root state is recurrent.

Theorem 3.1 (sharp tails)**.** *For every $n \geq 1$,*

$$\max_{T,r} \tau(T,r) = \left\lfloor \frac{n-1}{2} \right\rfloor. \tag{2}$$

Both parities admit explicit labelled caterpillar witnesses.

Proof. Suppose first that n is odd. After a move into an odd side of size s, the reverse edge points toward the complementary even side and is ineligible. Every next chosen odd side is a proper subset of the preceding one. The difference of two nested odd sizes is positive and even, so sizes fall by at least two. The first is at most $n-2$ and the last at least one, giving at most $(n-1)/2$ moves.

For even n, suppose a tail of length d reaches the two-cycle $v_d \leftrightarrow v_{d+1}$ along $v_0, v_1, \ldots, v_d, v_{d+1}$. For each $1 \leq i \leq d$, the vertex v_i has two distinct neighbours on this H-path. Its H-degree is odd, hence at least three, so it has an off-path H-neighbour. These d witnesses are distinct, since sharing one would make a cycle in H. Thus $n \geq (d+2)+d$ and $d \leq n/2-1$.

For odd $n = 2d+1$, take a spine $v_0 \cdots v_d$ and attach one leaf to each $v_0, \ldots, v_{d-1}$. Label each forward spine neighbour below the competing leaf; the root walks from v_0 to the fixed v_d. For even $n = 2d+2$, use the spine $v_0 \cdots v_{d+1}$ and attach one leaf to each $v_1, \ldots, v_d$. Every degree is odd, hence every edge lies in H; labels decreasing along the forward spine and larger on the leaves force a tail of d into $v_d \leftrightarrow v_{d+1}$. ☐

4. Exact recurrent enumeration

Let

$$R(z) = \sum_{n\geq 1} n^{n-1} \frac{z^n}{n!}, \qquad R(z) = z \exp R(z), \tag{3}$$

and split rooted trees by order parity:

$$O(z) = \frac{R(z)-R(-z)}{2}, \qquad E(z) = \frac{R(z)+R(-z)}{2}. \tag{4}$$

Theorem 4.1 (fixed and recurrent EGFs)**.** *For odd n, the number f_n of recurrent rooted states is*

$$f_n = n![z^n]\, z \exp(E(z)). \tag{5}$$

For even n, define

$$W(z) = z \exp(E(z)) \frac{\exp(O(z))-1}{O(z)}, \qquad W_{\text{odd}}(z) = \frac{W(z)-W(-z)}{2}. \tag{6}$$

Then the number r_n of recurrent rooted states is

$$r_n = n![z^n] W_{\text{odd}}(z)^2. \tag{7}$$

Hence the dynamical zeta function at fixed order is

$$\zeta_n(t) = \begin{cases} (1-t)^{-f_n}, & n \text{ odd}, \\ (1-t^2)^{-r_n/2}, & n \text{ even}. \end{cases} \tag{8}$$

Proof. At an odd-order fixed root every branch has even order. The labelled species $\mathsf{Z} \cdot \mathrm{SET}(\mathcal{T}_{\text{even}})$ gives (5).

For the even case, cut an oriented recurrent state $u \to v$ at its mutual edge. This produces an ordered pair of odd-order rooted trees A, B. If k_A is the number of odd-order branches at the root of A, then u chooses the cross-edge neighbour v precisely when the label of v is least among it and those k_A branch roots. Its relative-label probability is $1/(k_A+1)$. Marking odd branches by x and integrating gives the weighted rooted-tree EGF

$$\int_0^1 z\exp(E+xO)\,dx = z\exp(E)\frac{\exp(O)-1}{O} = W.$$

The analogous comparison on side B involves a disjoint label set, so the two relative-order events are independent. Both sides must have odd total order. Their ordered labelled product is therefore W_{odd}^2, proving (7). An orientation is a recurrent root state; the opposite orientation is its partner, so there are $r_n/2$ two-cycles. The zeta formula follows. □

The first values are

$$f_1, f_3, f_5, f_7 = 1, 6, 380, 68712, \qquad r_2, r_4, r_6, r_8 = 2, 56, 6512, 1718656.$$

5. Every-target fibres and their maximum

For adjacent u, v, let $C_u(v)$ retain the meaning from Section 1. An empty minimum below is interpreted as $+\infty$.

Theorem 5.1 (local inverse atlas)**.** *For every target (T, v),*

$$\operatorname{indeg}(T,v) = \mathbf{1}\{|C_v(w)| \text{ is even for every } w\sim v\} \tag{9}$$
$$+\ \#\left\{u\sim v:\ \begin{array}{l}|C_u(v)| \text{ is odd,}\\ v<\min\{w\sim u : w\neq v,\ |C_u(w)| \text{ is odd}\}\end{array}\right\}.$$

The sharp maximum over all targets and labelled trees is

$$\max_{T,v}\operatorname{indeg}(T,v) = \begin{cases}(n+1)/2, & n \text{ odd,}\\ n-1, & n \text{ even.}\end{cases} \tag{10}$$

Proof. The target is its own predecessor exactly under the displayed fixed-root condition. A different predecessor must have root at a neighbour u. Rule (1) maps it to v exactly when the v-side is odd and v is the least eligible neighbour of u, which is the set condition in (9). These possibilities are disjoint and exhaustive.

For even n there is no self predecessor, so the fibre is at most $\deg(v) \le n-1$. A star whose centre is the target attains $n-1$ because every leaf has only the centre as eligible neighbour. For odd n, every neighbour-source counted in (9) lies in an even-order branch at v. There are at most $(n-1)/2$ such branches; adding the possible self predecessor gives $(n+1)/2$. This is attained by a fixed centre joined to $(n-1)/2$ paths of length two, with the centre given the least label. Each branch has order two, and its inner vertex maps back to the centre. □

Longer fibres are counted without ambiguity by iterating the local reverse test in (9); because the carrier tree is fixed, this is a finite dynamic program on its oriented edges and vertices.

6. Controls and limitations

The paper-local verifier generates every labelled tree from its Prüfer code through $n = 8$, every distinguished root, and every transition. It checks the parity classifications, direct orbit depths and periods, sharp maxima, the local fibre formula, both EGF coefficient families, and complete depth histograms. This is regression pressure, not proof or novelty evidence.

The map deliberately keeps the tree fixed. We do not claim results for unlabelled trees, dynamically changing labels, stochastic tie breaking, general graphs, or weighted side congruences. Most of the species algebra is classical after the recurrent structures are identified. External posting and submission remain unauthorized under `OWNER_AMBER/HOLD_EXTERNAL`.

References

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